Identification of specific autoantibody targets in Alzheimer's Disease

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INTRODUCTION AND OBJECTIVES

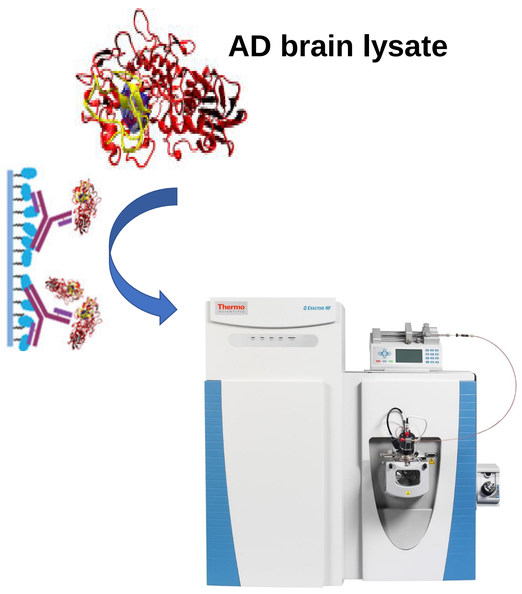
Alzheimer's disease (AD) is a progressive and chronic neurodegenerative disorder that affects wide areas of the cerebral cortex and hippocampus. It is the most common form of dementia worldwide, with a 10-30% prevalence in the ageing population (>65 years of age), a mean incidence of 1-3% and a high socioeconomic impact. The clinical phases are preceded by preclinical and prodromal stages, which are extended over 2-3 decades before the first symptoms. AD patients' brains are characterized by two main hallmarks: the formation of intracellular neurofibrillary tangles of hyperphosphorylated Tau and the presence of extracellular amyloid plaques. The specific detection of AD, for which definitive diagnosis requires *post mortem* verification, includes molecular PET imaging, measurement of Aβ and tau in the cerebrospinal fluid and behavior history assessments. However, these methods are not enough for early detection. Therefore, new diagnostic biomarkers and therapeutic targets of intervention are necessary.

Due to the drastic changes in AD brains caused by neurodegeneration, the blood brain barrier is altered and the brain enters into a permanent inflammatory state. These may lead to the generation of **AD-specific autoantibodies.**

Our objective is the identification of their target proteins, which might become a new source of AD biomarkers and a new approach to study AD preclinical and clinical phases at the cellular and molecular levels.

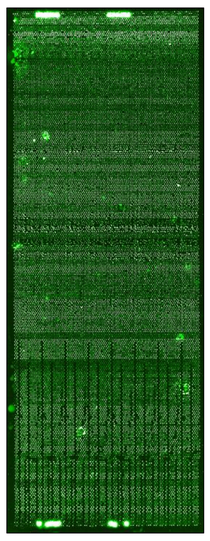
GENERAL WORKFLOW AND TECHNIQUES

AD brain lysate

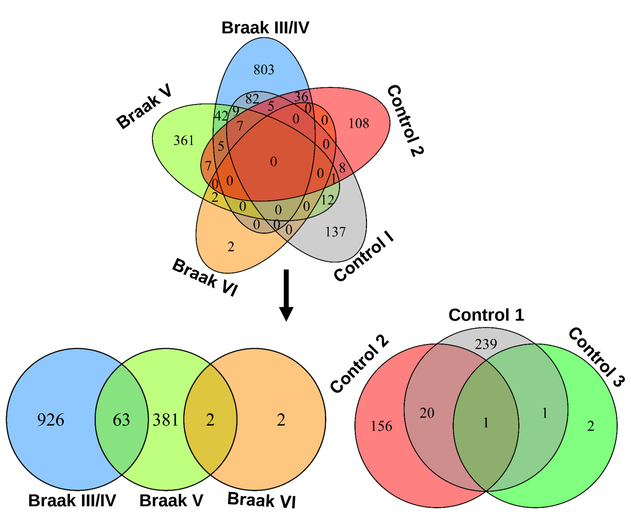


IP-LC MS/MS analysis
(15 candidate targets in 40 PrESTs)

12 x 21K-arrays
3 different AD pools and 2 different control pools (4 inidivual serum samples/each)



Analysis of the results

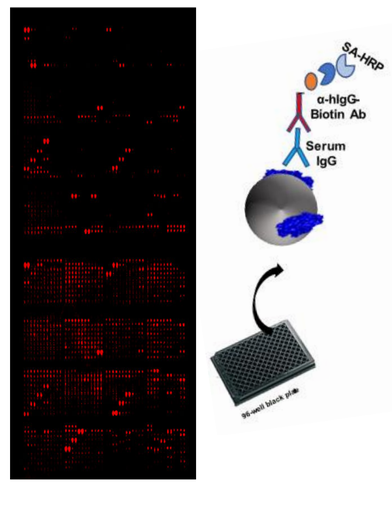


42K ARRAYS (196 PrESTs corresponding to 196 proteins)

21 equal subarrays containing 384 PrESTs
(x2 arrays→ 40 serum samples + control)
x5 different arrays (~1920 PrESTs)

384 PrEST ARRAYS (23 selected PrESTs)

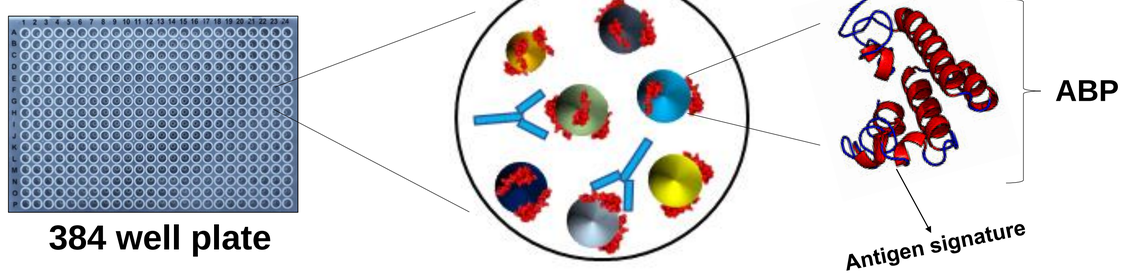
2 different T7 Phage libraries
1920 phages and controls in duplicate per array



PHAGE ARRAYS
(6 targets in 13 PrESTs)

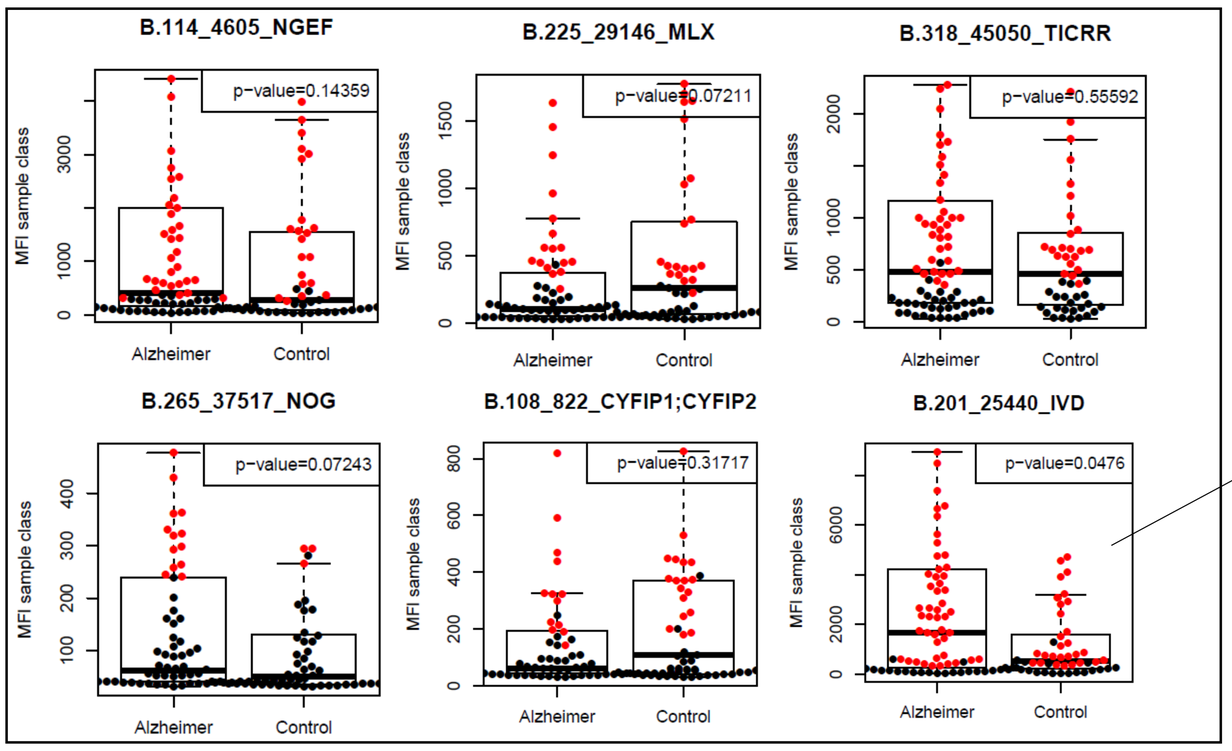
IgG-REACTIVITY ALIDATION USING 384 BEADS PrEST ARRAYS

384 BEADS ARRAY
380 PrESTs + 4 controls



384 well plate

128 serum samples
(77 AD + 59 controls)



Conclusions

We have performed four different high-throughput proteomic approached to describe the humoral response of Alzheimer's Disease patients.

More than 100 potential new targets of AD-specific autoantibodies have been identified using these approached.

Validation of the results using a 384 PrEST beads array and 128 serum samples (77 AD patients and 59 healthy individuals) shows IVD is specifically recognized by AD patients (Mann-Whitney U test p value > 0.05)

Future prospects

To confirm the usefulness of these proteins as specific AD biomarkers, we will use a higher cohort of human sera samples, including healthy individuals and patients with other dementias.

We will also analyze if these proteins show altered expression in different brain regions, especially those early affected in the course of the disease, and if the play an important functional role in AD pathogenesis.